

基於 Java Servlet、JSP 和 MVC 設計模式 的 圖書管理系統，其中 BookDAO 改為使用 IO 序列化 (Serialization) 存取資料。

✧ 功能概述

Book (Model)：代表一本書，包含 id、bookname、price、author、publisher，並實作 Serializable 介面，以支援 物件序列化。

BookDAO：使用 檔案 (File) 讀寫，透過 ObjectOutputStream / ObjectInputStream 來序列化及反序列化 Book 物件清單。

BookController (Servlet)：負責處理請求 (新增、刪除、修改)。

JSP 頁面 (View)：提供 CRUD 介面。

✧ 程式結構

/BookApp

```
| —— src/  
|   |—— model/  
|   |   |—— Book.java  
|   |   |—— BookDAO.java  
|   |—— controller/  
|   |   |—— BookController.java  
| —— webapp/  
|   |—— index.jsp  
|   |—— addBook.jsp  
|   |—— editBook.jsp  
|   |—— listBooks.jsp  
| —— web.xml  
| —— books.dat (書籍資料檔案)
```

1 建立 Book 類別 (Model - 可序列化)

```
package model;
```

```
import java.io.Serializable;
```

```
public class Book implements Serializable {  
    private static final long serialVersionUID = 1L; // 確保版本一致  
  
    private int id;  
    private String bookname;  
    private double price;  
    private String author;
```

```

private String publisher;

public Book() {}

public Book(int id, String bookname, double price, String author, String
publisher) {
    this.id = id;
    this.bookname = bookname;
    this.price = price;
    this.author = author;
    this.publisher = publisher;
}

public int getId() { return id; }
public void setId(int id) { this.id = id; }

public String getBookname() { return bookname; }
public void setBookname(String bookname) { this.bookname = bookname; }

public double getPrice() { return price; }
public void setPrice(double price) { this.price = price; }

public String getAuthor() { return author; }
public void setAuthor(String author) { this.author = author; }

public String getPublisher() { return publisher; }
public void setPublisher(String publisher) { this.publisher = publisher; }

@Override
public String toString() {
    return "Book [id=" + id + ", bookname=" + bookname + ", price=" + price +
        ", author=" + author + ", publisher=" + publisher + "]";
}
}

```

◆ **Serializable** 允許 **Book** 物件被存入 **books.dat**，方便永久儲存與讀取。

2 BookDAO - 使用 IO 讀寫

```
package model;

import java.io.*;
import java.util.ArrayList;
import java.util.List;

public class BookDAO {
    private static final String FILE_NAME = "books.dat";
    private List<Book> bookList;

    public BookDAO() {
    }

    // 讀取 `books.dat` 檔案
    private List<Book> loadBooks() {

    }

    // 儲存 `books.dat` 檔案
    private void saveBooks() {

    }

    // 取得所有書籍
    public List<Book> getAllBooks() {
    }

    // 新增書籍
    public void addBook(Book book) {

    }

    // 搜尋書籍
    public Book findBookByName(String name) {

    }
}
```

```
// 刪除書籍
    public void deleteBook(int id) {

    }

// 修改書籍
    public void updateBook(int id,Book book) {

    }

}
```

◆ 透過 `ObjectOutputStream` 和 `ObjectInputStream` 存取 `books.dat`，實現序列化 (Serialization) 存取。

3 BookController (Servlet - 處理前端請求)

```
package controller;

import model.Book;
import model.BookDAO;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.IOException;

@WebServlet("/BookController")
public class BookController extends HttpServlet {
    private static final long serialVersionUID = 1L;
    private BookDAO bookDAO = new BookDAO();

    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
        String action = request.getParameter("action");

        if ("add".equals(action)) {
```

```

        } else if ("delete".equals(action)) {

        } else if ("update".equals(action)) {

        }

        response.sendRedirect("listBooks.jsp");
    }
}

```

4 JSP 頁面 (View)

✎ listBooks.jsp

```

<table border="1">
    <tr><th>ID</th><th>書名</th><th>價格</th><th>作者</th><th>出版社
</th><th>操作</th></tr>

</table>

```

IO (books.dat) 儲存書籍，實現物件序列化 (Serialization)。

Book (Model)：可序列化 (Serializable)

BookDAO：使用 IO (ObjectOutputStream / ObjectInputStream)

BookController：處理請求

JSP (View)：顯示書籍資訊